

Req ID	Type	Description	Priority	Acceptance criteria	Rationale	Comments
FR-001	Functional	The system shall generate optimized daily collection routes including only bins that have exceeded an 80% fill threshold.	High	Pass: Route excludes bins < 80% full unless manually tagged Fail: Overflowing bin omitted from driver manifest.		
FR-002	Functional	System should allow driver to mark a bin as "collected" or "missed" along with timings	High	Pass: Each status change records timestamp and GPS location Fail: Status update missing timestamp or location data.	Allows driver to keep track of bins	
FR-003	Functional	System shall have gps routing to each bin in route	High	Pass: Driver app loads route Fail: Navigation fails to load or bin coordinates are incorrect.	Driver should be able to navigate to target bins within the same interface	
FR-004	Functional	The system shall automatically reassign uncollected bins from missed-pickup trucks to nearby alternative trucks within 10 minutes of "missed" status.	High	Pass: Reassigned truck receives updated route with new bins within 10 min; Fail: Reassignment delayed >15 min or new truck not notified.	Will ensure that minimal / no bins are missed	

FR-005	Functional	The system shall generate daily compliance reports (e.g., % of scheduled bins collected, missed-pickup reasons, safety incidents).	Medium	Pass: Report email received with complete data for that day; Fail: Report missing, delayed, or incomplete.		
NFR-001	Non-functional	Route recalculation for a fleet of 50 trucks across 10,000 bins shall execute in under 45 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.		
NFR - 003	Non-functional	System interface should reflect updates on bins collected/missed in interface within 2 seconds of update	High	Pass: updates on bin collection status reflected to drivers in <2 seconds Fail: update takes longer to reflect/ doesn't reflect	Will allow better real time syncing and coordination	

USE-CASE FLOW SPECIFICATION

Use Case: Mark Bin as Collected/Missed

Main Success Scenario:

1. The truck driver navigates to the bin location using GPS.
2. The driver physically collects garbage from the bin or verifies bin condition.
3. The driver opens the mobile app and selects the "Mark Collected" or "Mark Missed" button.
4. The system automatically captures GPS coordinates and current timestamp.
5. The system validates that the bin is on the driver's assigned route for the day, and that bin is now empty using smart sensing.
6. The system updates the bin status in the database to "Collected" or "Missed".
7. If "Collected," the system updates the remaining bin count.
8. The system displays a green confirmation to the driver.
9. The driver acknowledges and proceeds to the next bin on the route.
10. Use case ends successfully.

Alternate Flow:

Bin is Inaccessible or Already Collected

1. Driver finds the bin location inaccessible (locked gate, damaged bin, or already collected). The driver taps the "Mark Missed" button instead.
2. The system displays a dropdown menu with predefined reasons: "Inaccessible," "Damaged," "Already Collected," "Other."
3. The driver selects a reason (e.g., "Inaccessible").
4. The system captures GPS, timestamp, and reason code in the database.
5. System sends a real-time alert to the Sanitation Supervisor within 2 minutes: "Bin #1047 MISSED Inaccessible (Driver: nala, Time: 08:52)."
6. System displays an amber confirmation message: "Missed pickup recorded. Supervisor notified."
7. The driver acknowledges and continues to the next bin. Use case ends (missed pickup escalates to Reassign Bins).